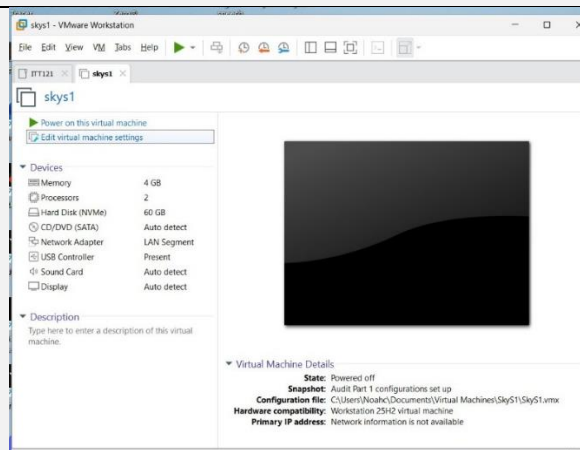
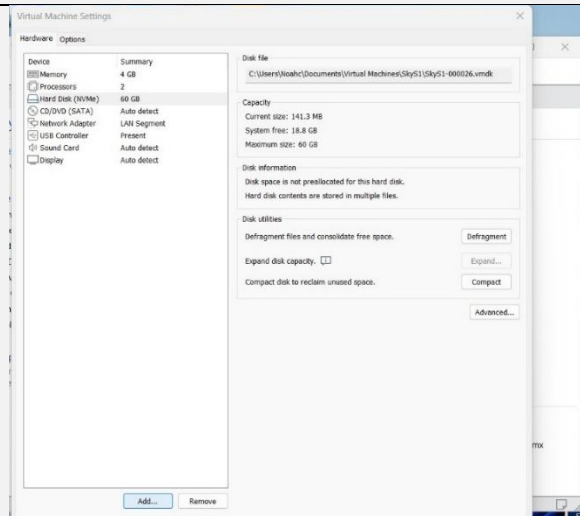


Backup and Restore

To start this assignment we will only need our server. Before we start the server we need to add a new hard disk for our backup. The VM needs to be powered off. When the VM is off find the “edit virtual machine settings”

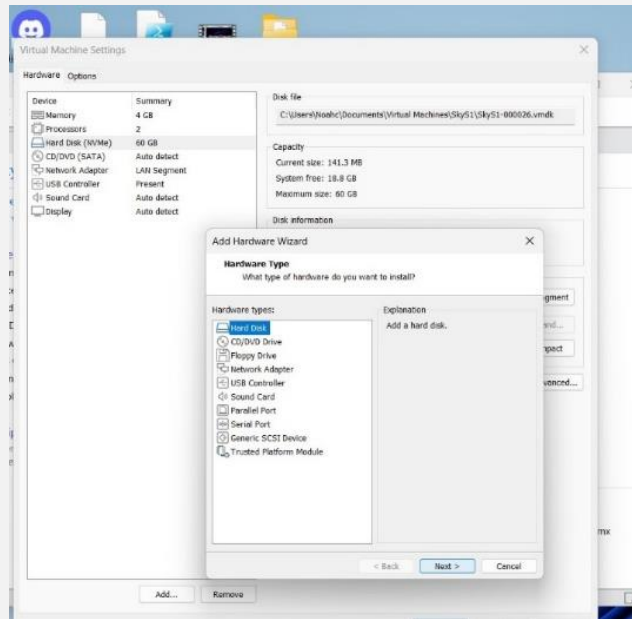


Inside the new window click the add option at the bottom of the window.

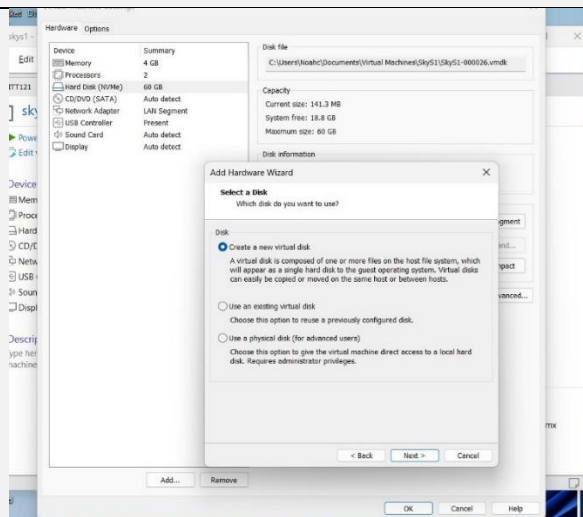


Backup and Restore

Select Hard Disk and then click next.

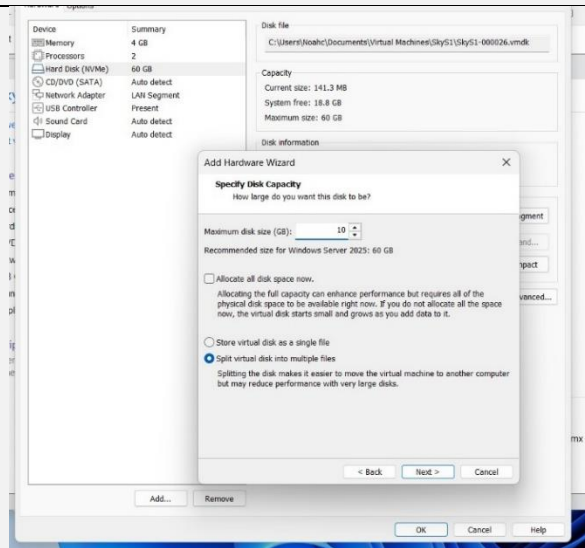


Keep the option, “create a new virtual disk”, and hit next.

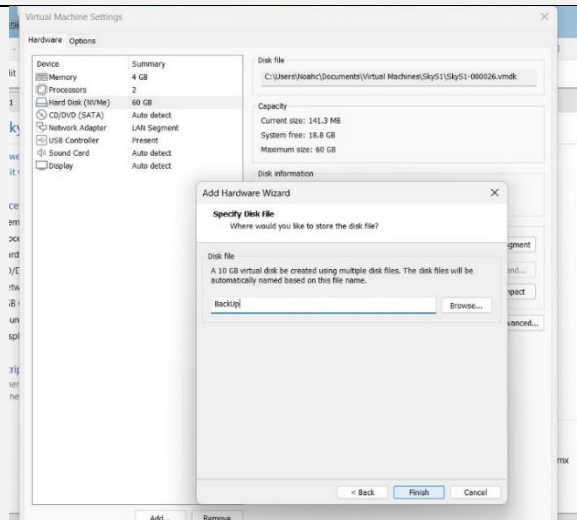


Backup and Restore

For the disk size make it 25GB so that it is big enough for all that we need to backup. Also keep the radio box the same and hit next.

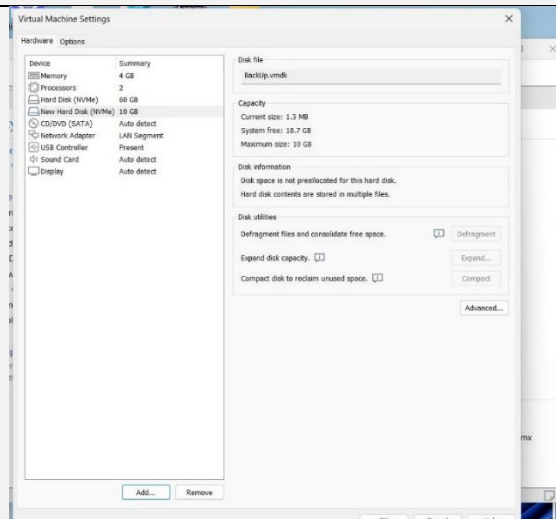


Give the new disk the name BackUp and click finish.

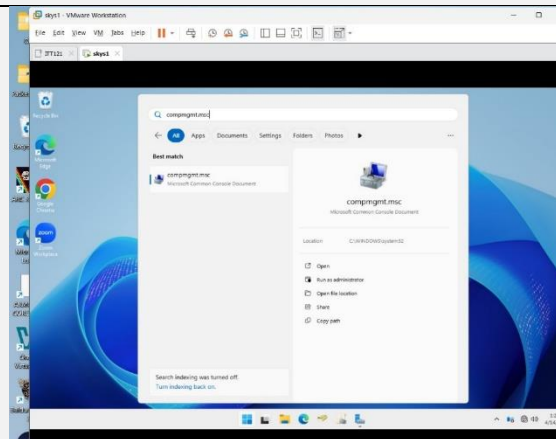


Backup and Restore

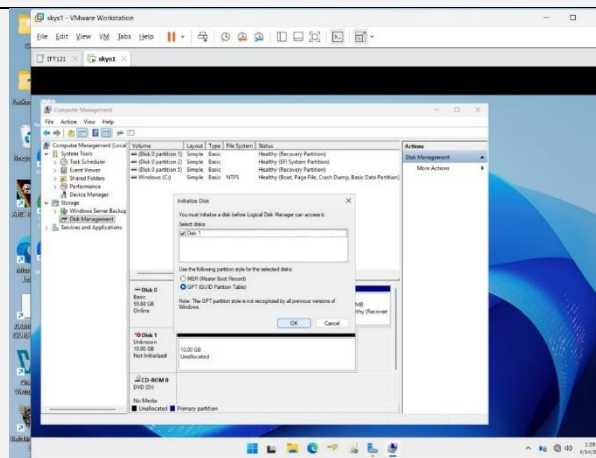
In the VM settings you should see your new BackUp disk alongside your main disk.



Now we can start the server VM. Once inside and logged in, click the windows icon (run menu) and type in “compmgmt.msc” and hit enter.

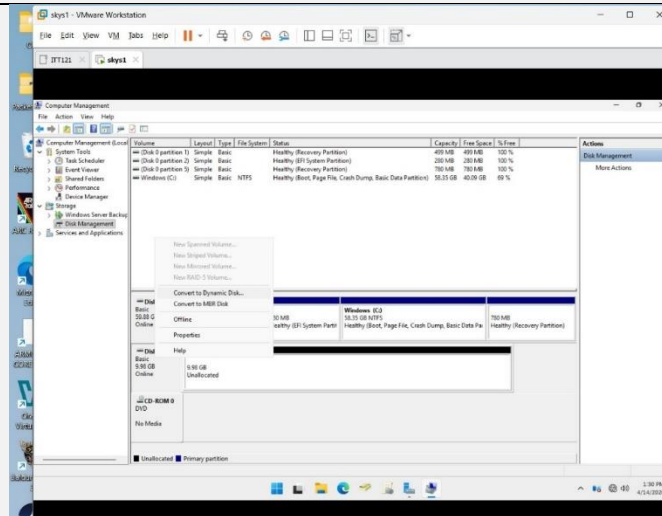


In the list to the left, find disk management and click on it. When you do click ok on the box that pops up. And make sure the Disk is selected.

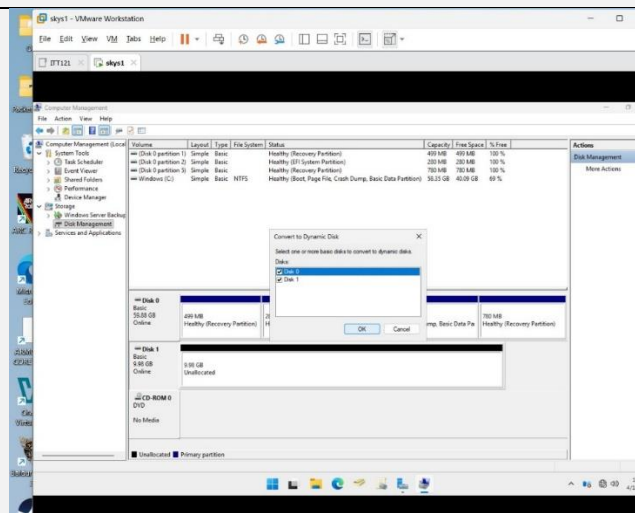


Backup and Restore

Now right click the new disk we made and choose the option, “convert to dynamic disk”.

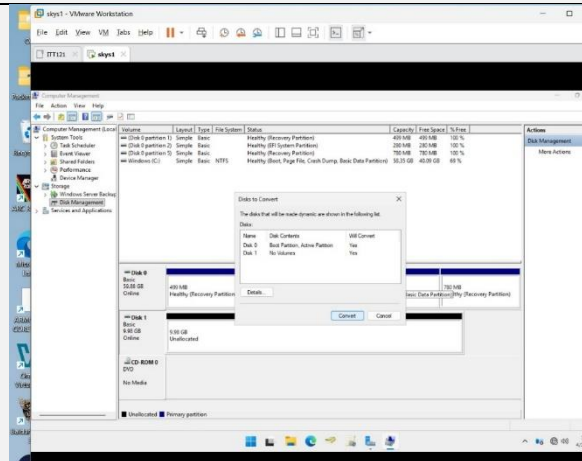


Select all the disk options and click ok.

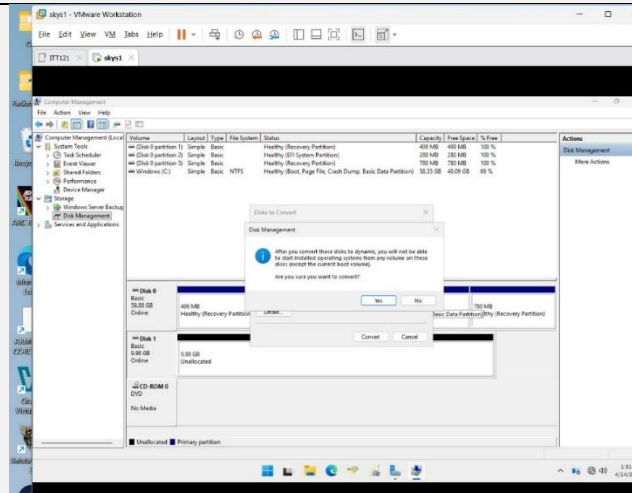


Backup and Restore

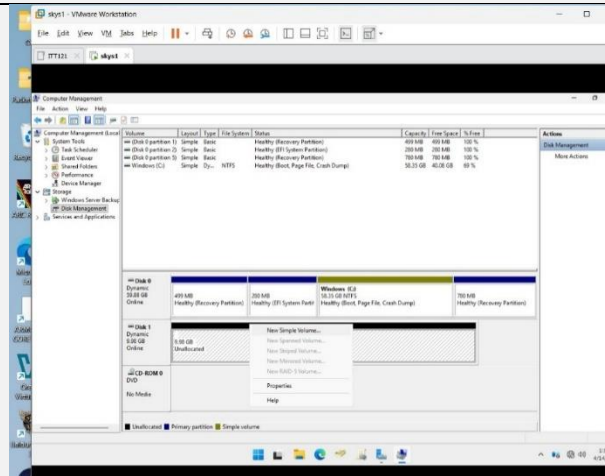
Now click the convert button.



Click ok on the information box that comes up.

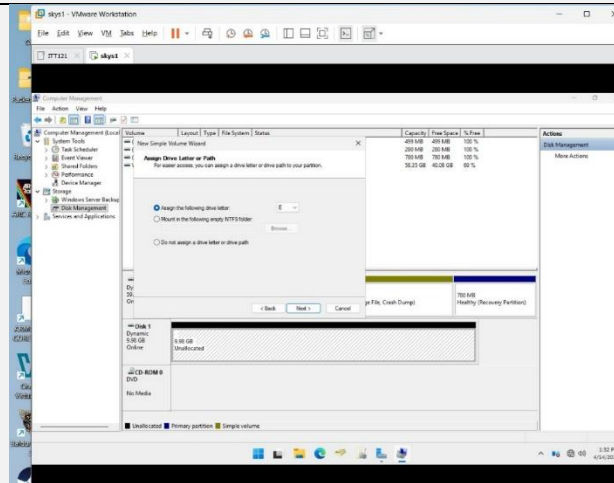


Now both boxes will have been converted. Right click on the black bar in the new disk we created and select "new simple volume"

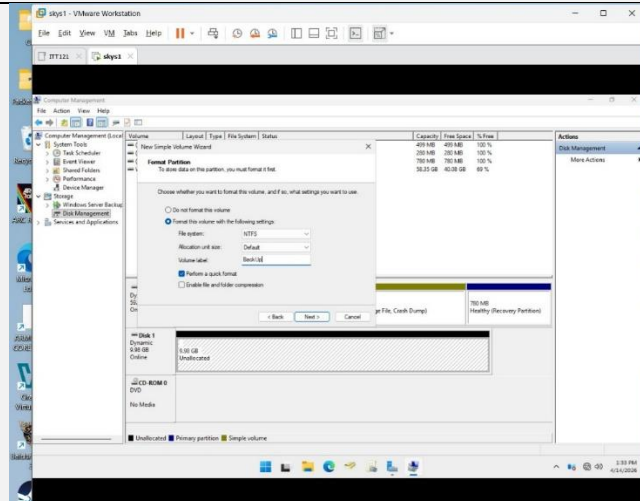


Backup and Restore

Continue through the pages, assign a the next available drive letter you have (E or F) and click next.

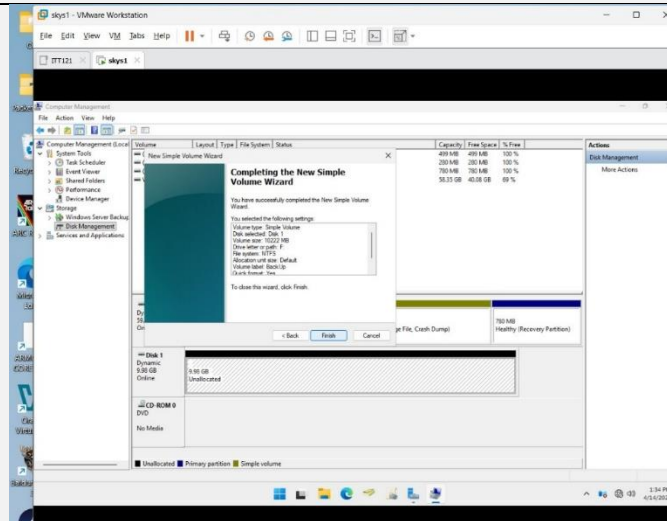


Continue until you reach this page and change the volume label to “Backup” and click next.

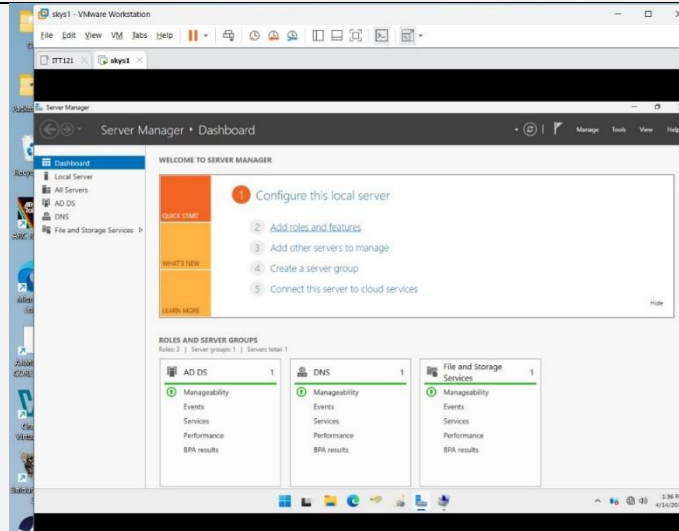


Backup and Restore

Click finish.

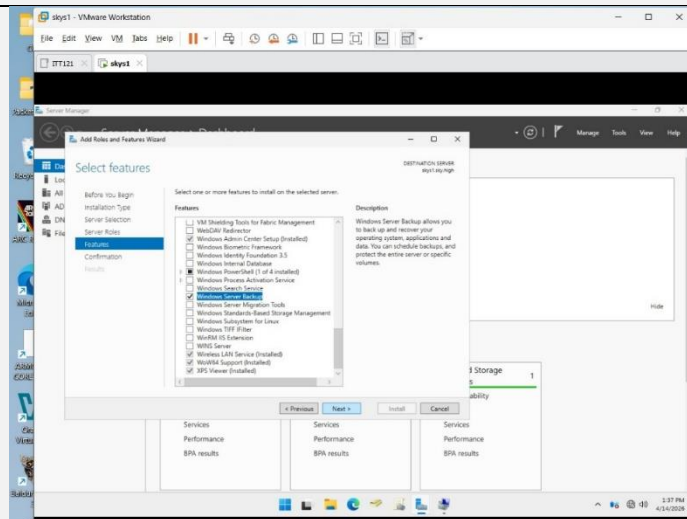


Now we need to add the local backup feature. Go to the server manager and click on “add roles and features”

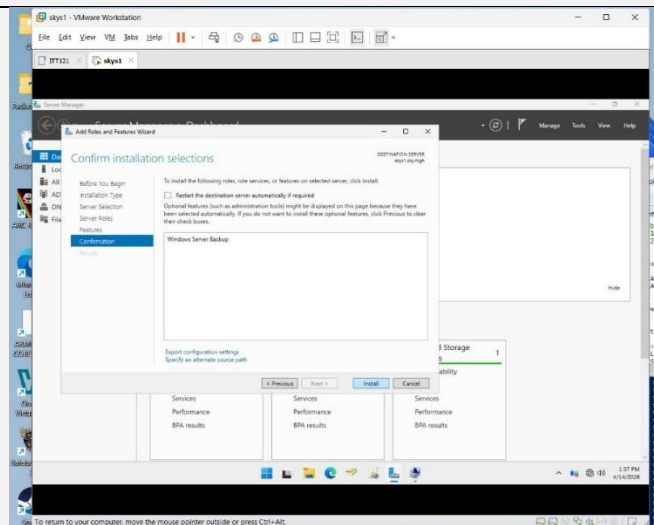


Backup and Restore

Click next through the pages until you reach the select features, scroll down until you find the “Windows Server Backup”. When you find it select it and click next.

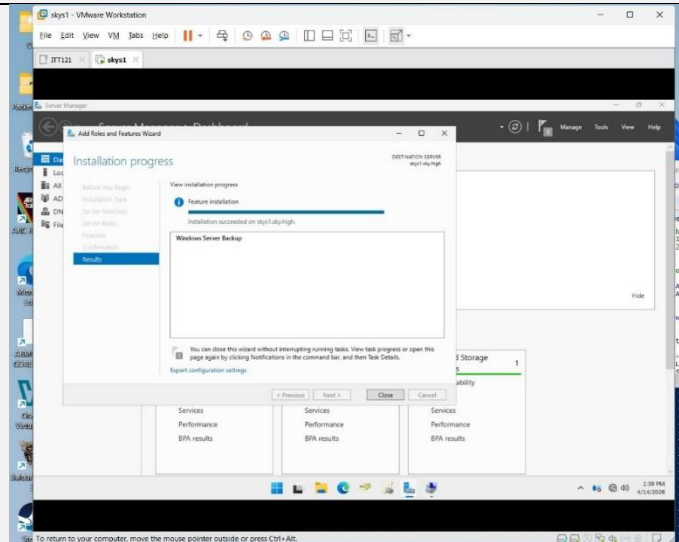


Click through the pages until you reach the install option, click install.

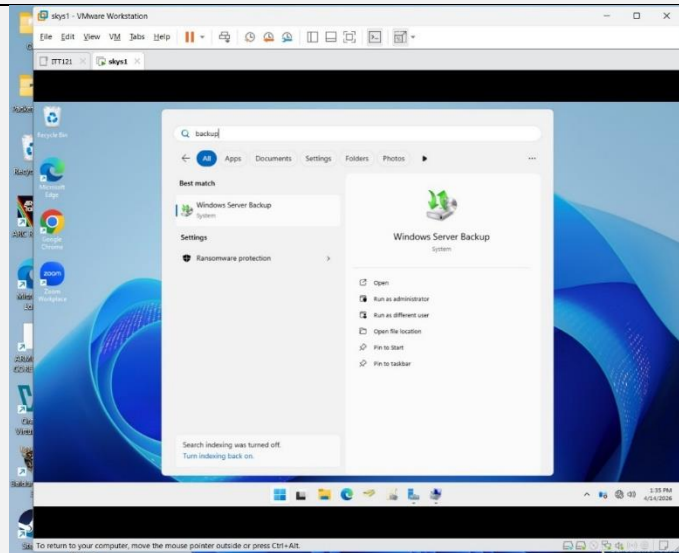


Backup and Restore

When the install is done you can close the application.

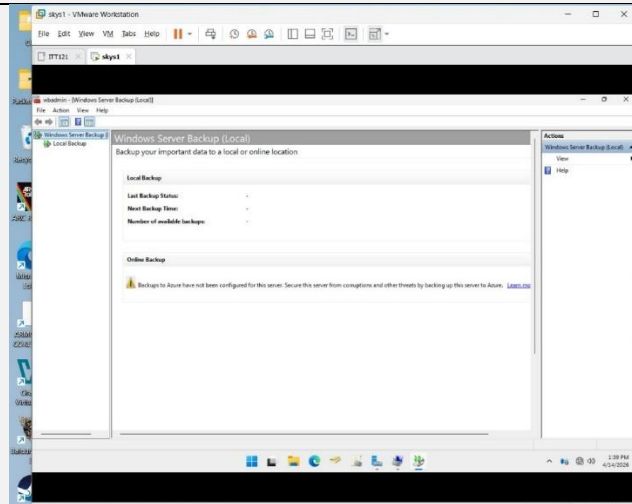


Now in the run menu, which you can access by clicking the windows icon, type out “backup” and hit enter.

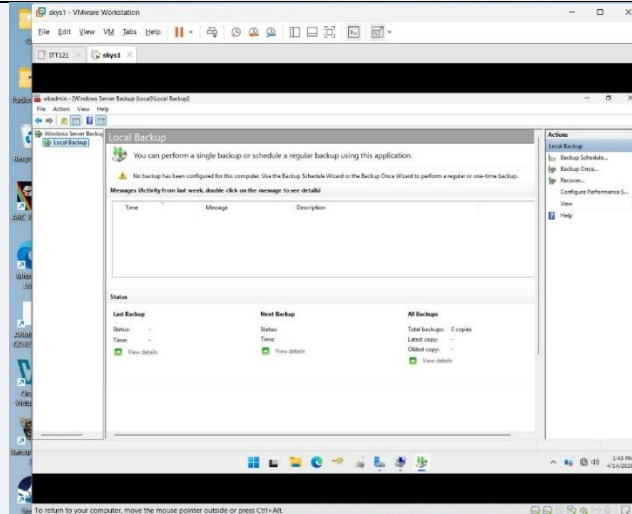


Backup and Restore

Now in the main page and on the left hand side, click the local backups.

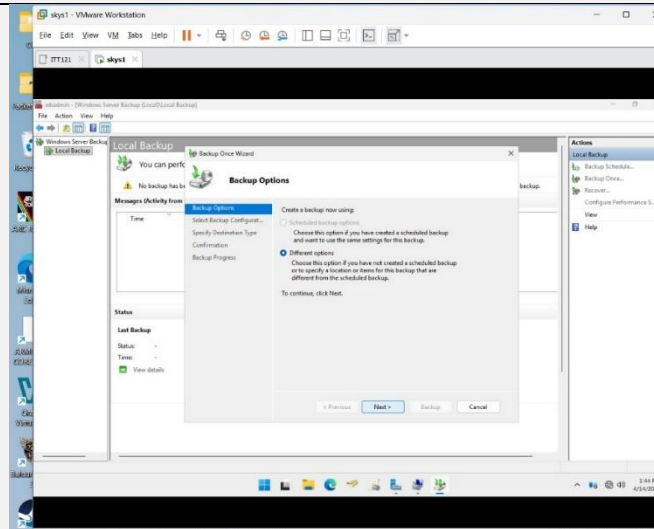


Once you are in local backups, on the right hand side click “Backup once”

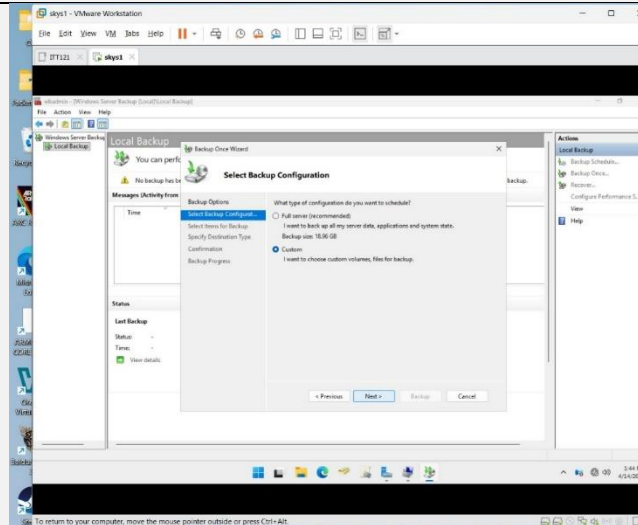


Backup and Restore

On the first page make sure you choose “different options” and hit next.

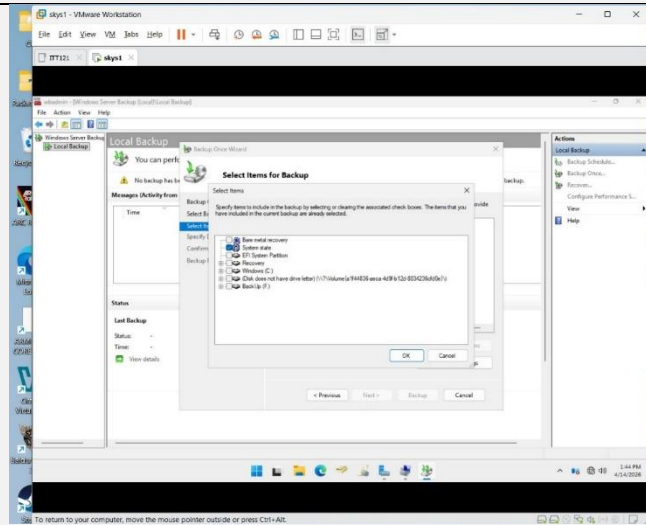


On the next page select “custom” and then click next.

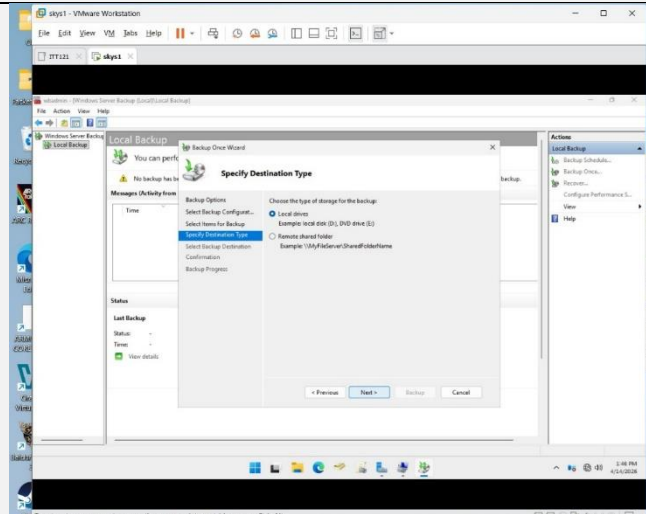


Backup and Restore

Now select “System State”
and click ok.

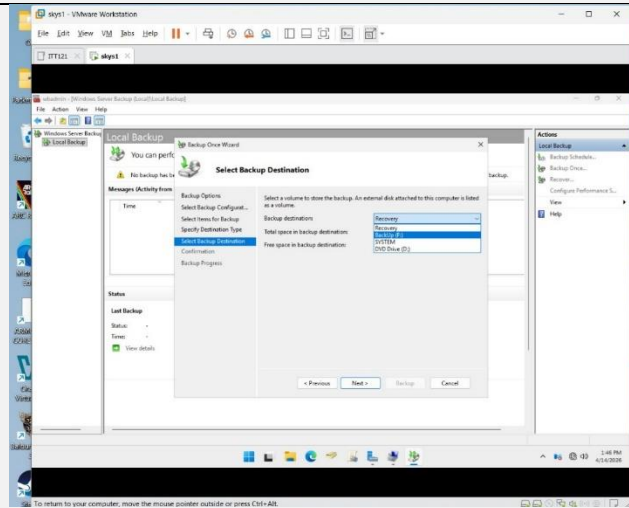


Make sure that local drives is selected and then click next.

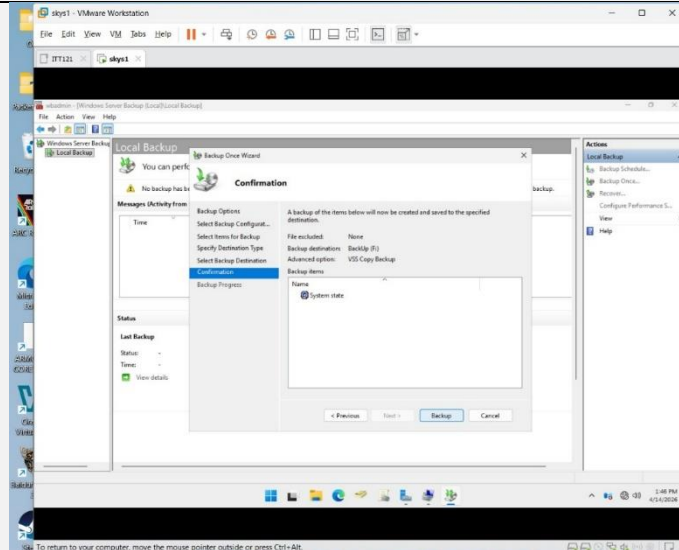


Backup and Restore

In the drop down box select Backups which should also have the drive letter you assigned to it. Click next after choosing your backup from the drop down box.

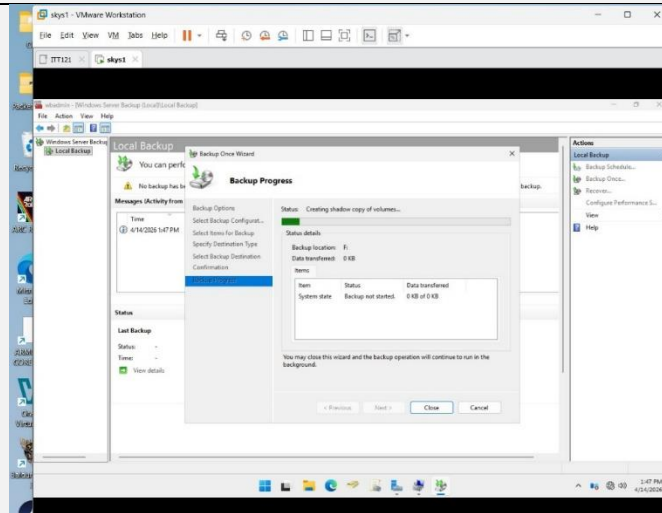


Now you can click “backup”.

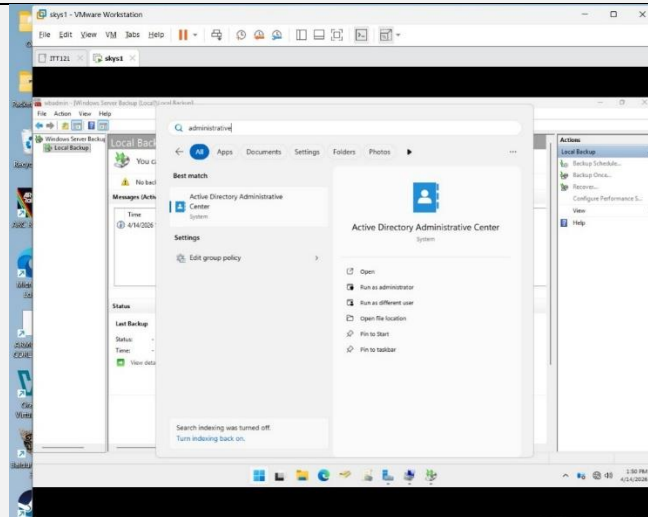


Backup and Restore

This may take a while to backup but you can see the progress.

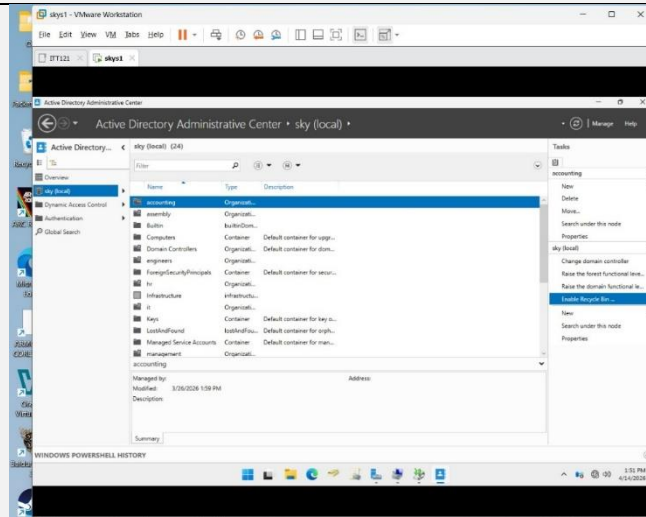


While we wait for the back up we also want to enable the recycling bin. In the start menu type out “Administrative Center” and look for the Active Directory Administrative Center.

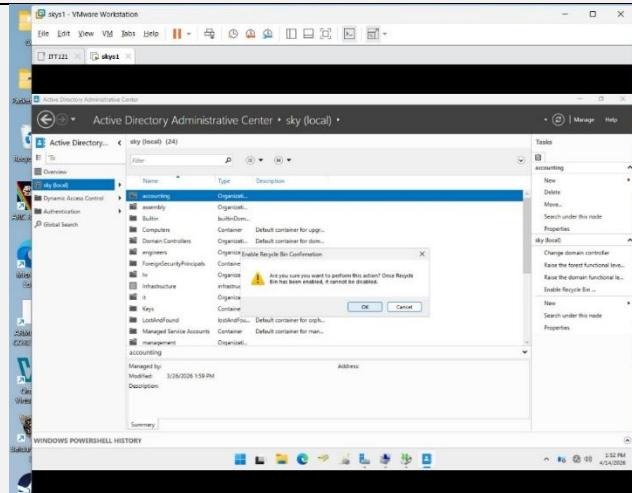


Backup and Restore

Open Active Directory Administrative Center and on the right hand side click “Enable Recycle Bin”

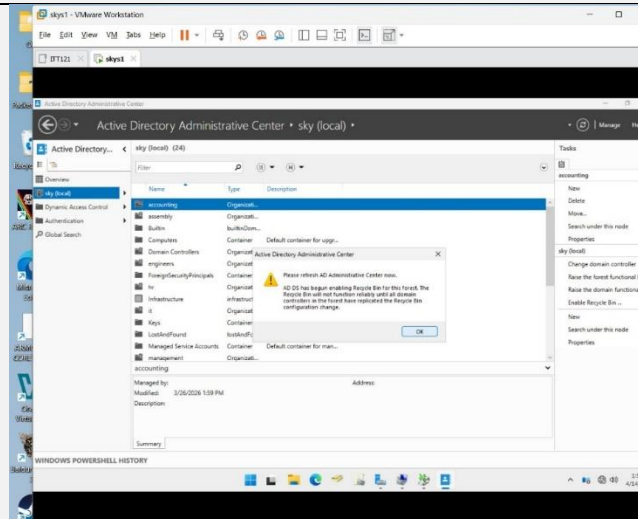


You will get a warning box and you can click ok on it.

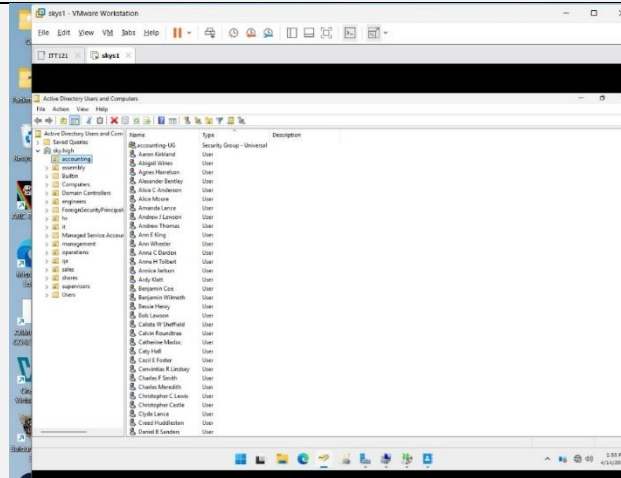


Backup and Restore

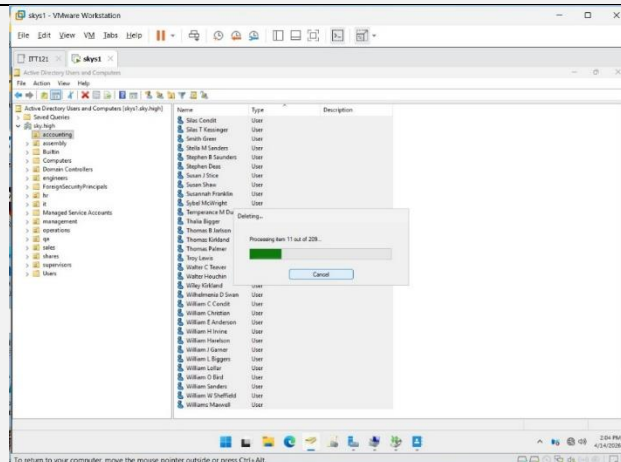
You will get another warning box and you can click ok on this too.



Now open Active Directory Users and Computers. You may want to take a snapshot before we move forward.

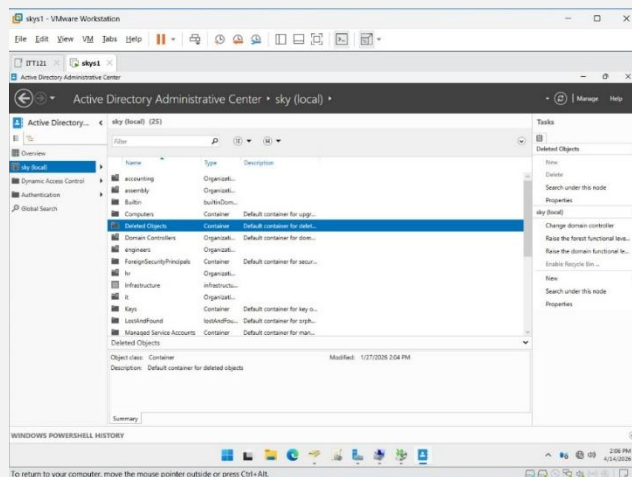


Go through all of your departments and delete all the users in the department folders.

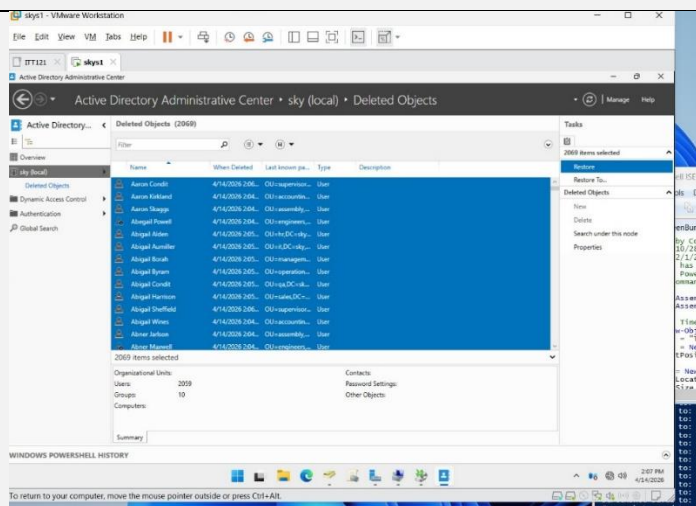


Backup and Restore

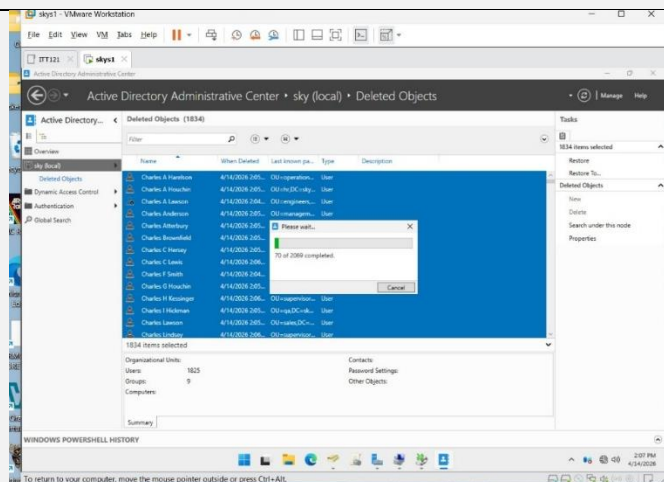
Once all of the users are deleted, go back to Active Directory Administrative Center. On the left hand side click on the local option. In the local list, find “deleted objects”.



Select everything in deleted objects, then click restore on the right hand side.

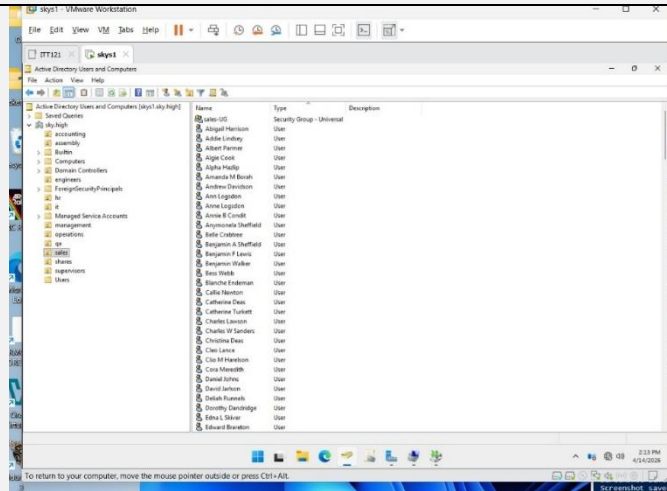


You will now see all of the users we deleted being restored.

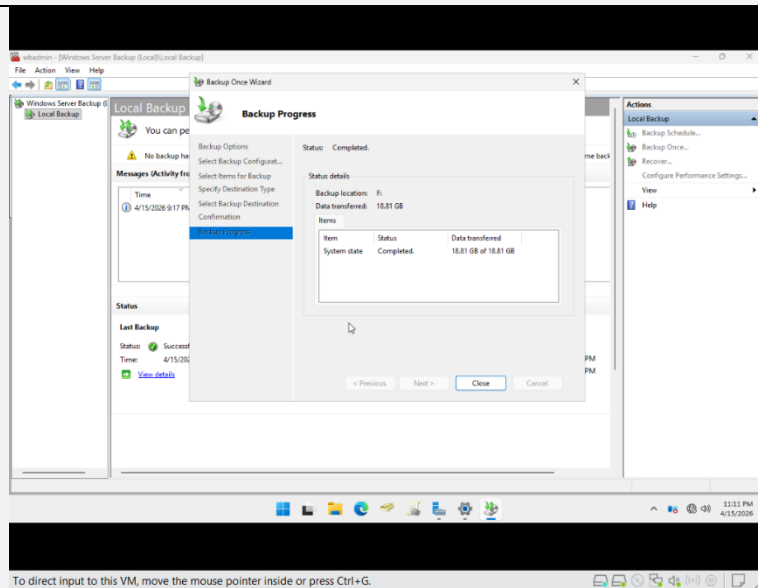


Backup and Restore

If we go back to Active Directory Users and Computers, all of users are back in the department folders.

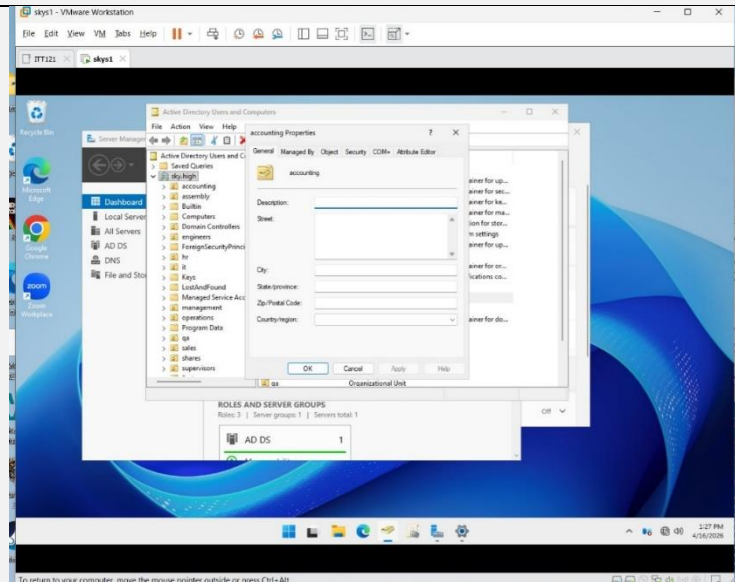


Now we can check the backup progress and if it is done we can click close.

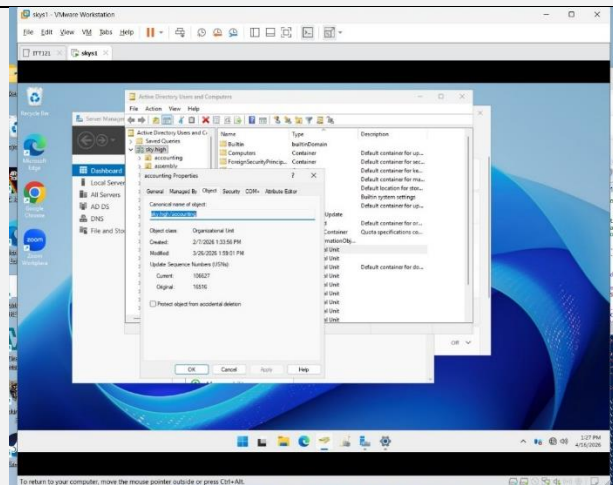


Backup and Restore

Now we are going to test our backup by deleting a department folder. In Active Directory Users and Computers, right click on accounting and delete the folder. You may need to go into advanced view and disable the prevent deletion for the department folder. You can do this by clicking on advanced under the view option in the top toolbar.

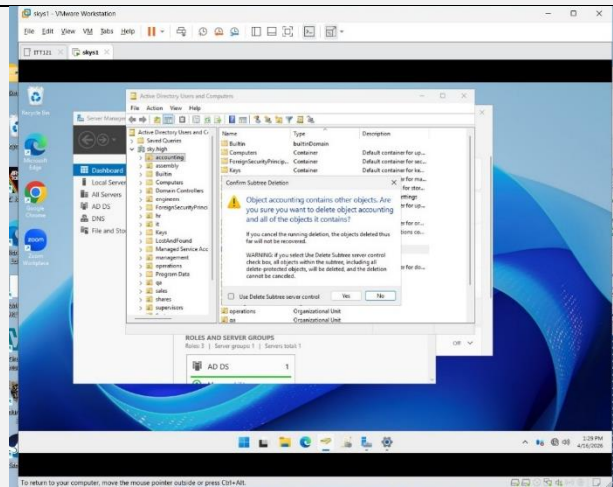


Make sure that you deselect the “prevent from accidental deletion” if needed.

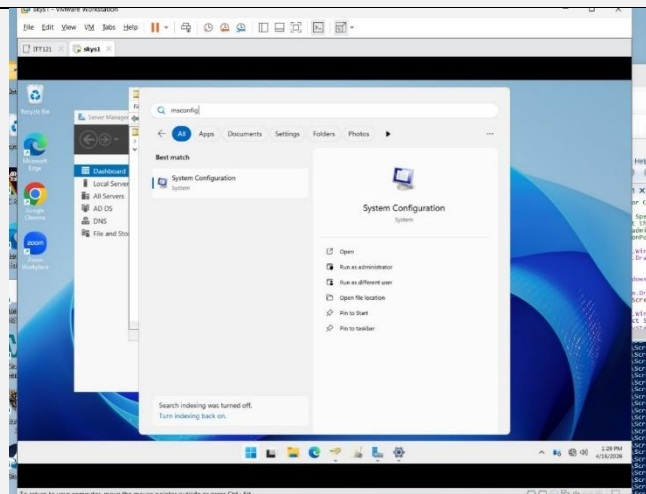


Backup and Restore

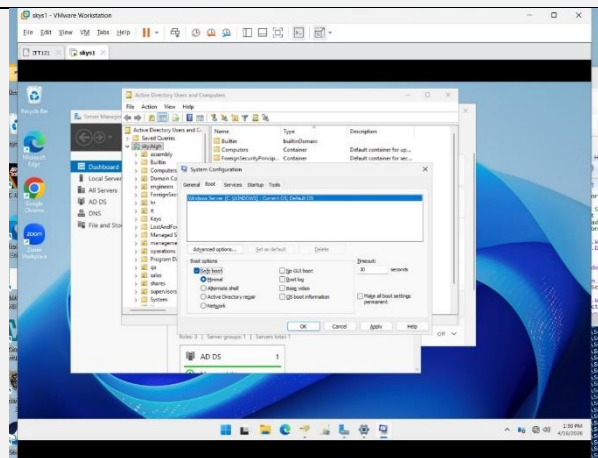
We should get a warning when we try to delete the folder, we can click yes.



Now we want to relaunch in safe mode. So in the start menu, search “msconfig” and hit enter.

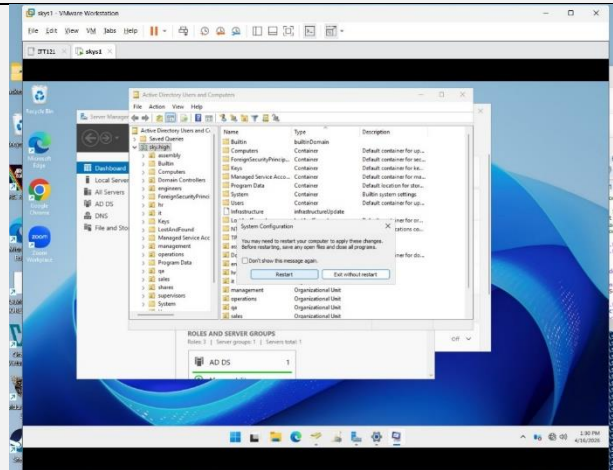


Go to the boot tab and at the bottom, select safe mode and minimal. Click apply and ok.

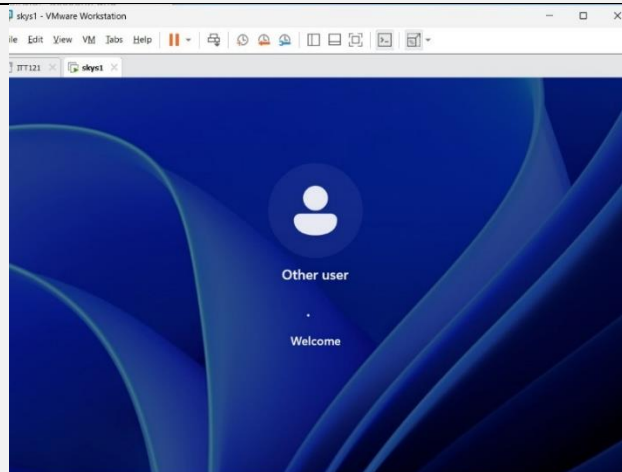


Backup and Restore

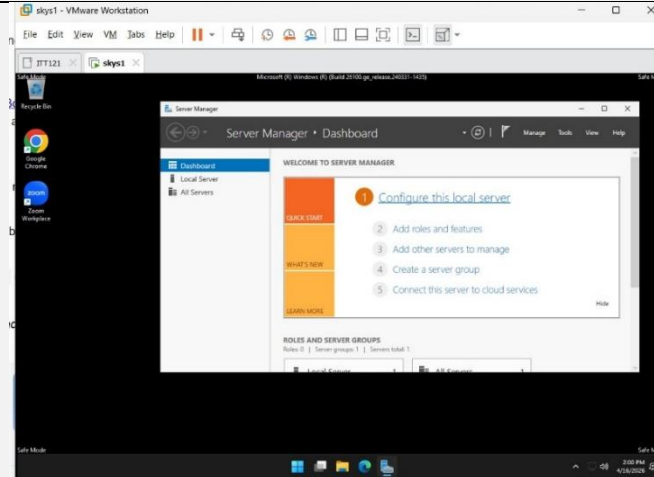
We will be prompted to restart, we can click restart.



Once the machine restarts log in as the local administrator.

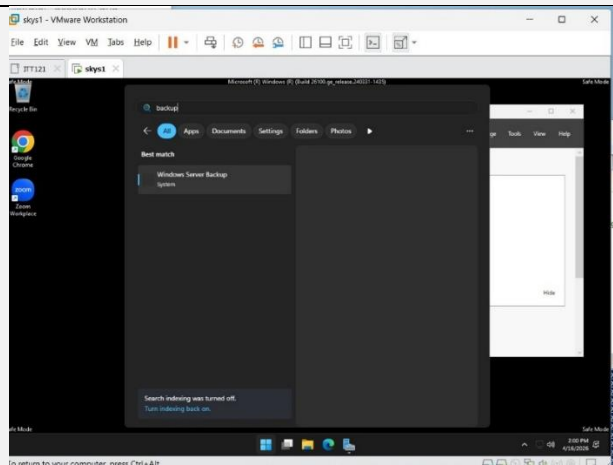


You should notice that we are in safe mode because it will say it at the bottom left and the background is different.

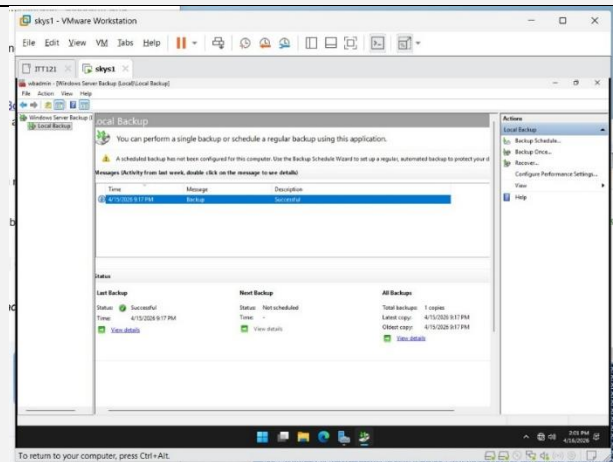


Backup and Restore

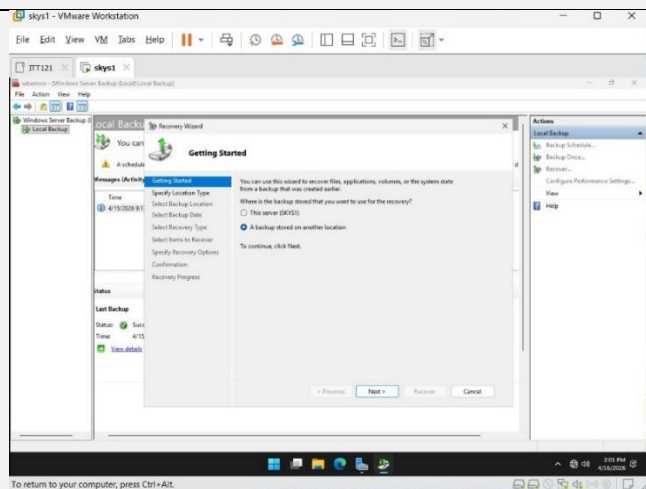
In the start menu search up
“backup” and hit enter.



In the local backups, click
on the backup we just made
and on the right hand side,
click “Recover”.

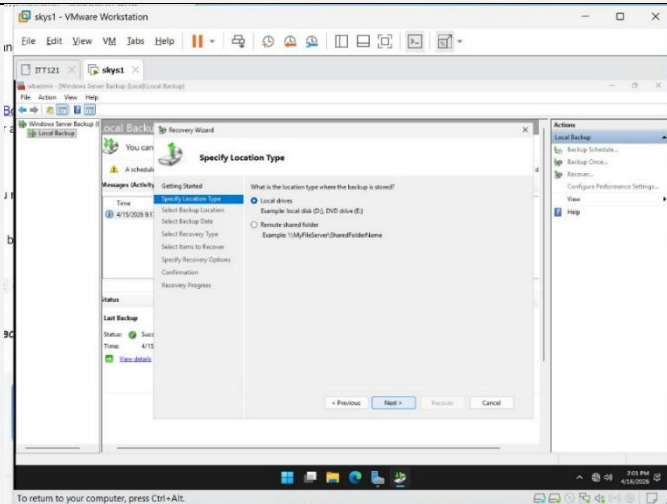


Make sure the “A backup
stored on another location”
is selected and click next.

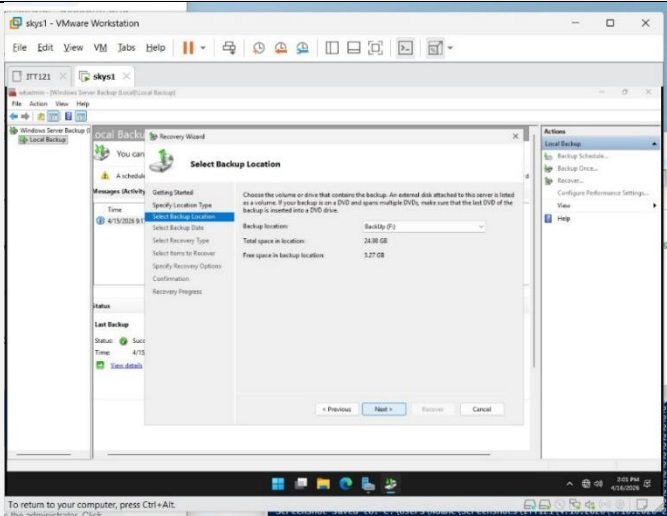


Backup and Restore

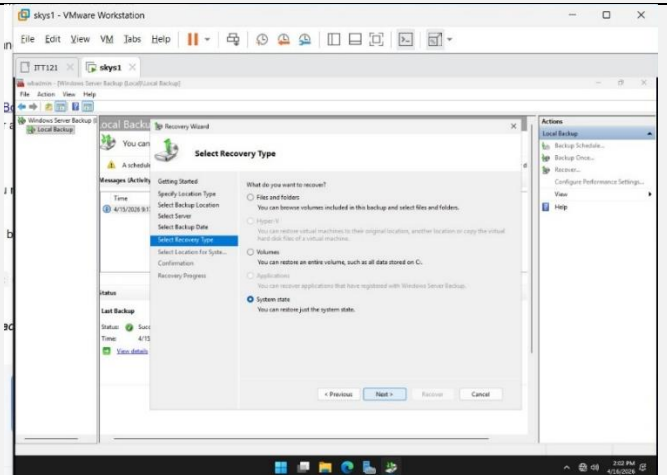
In the next page select the “Local drives” option and click next.



In the drop down menu find your backup drive and select it. Click next.

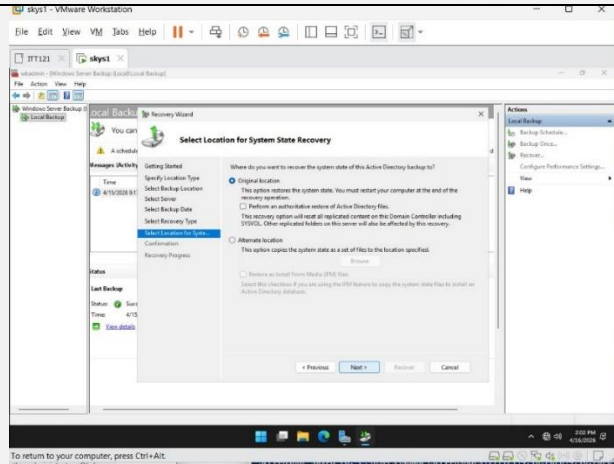


In the select recovery type page, make sure that system state is selected. Click next.

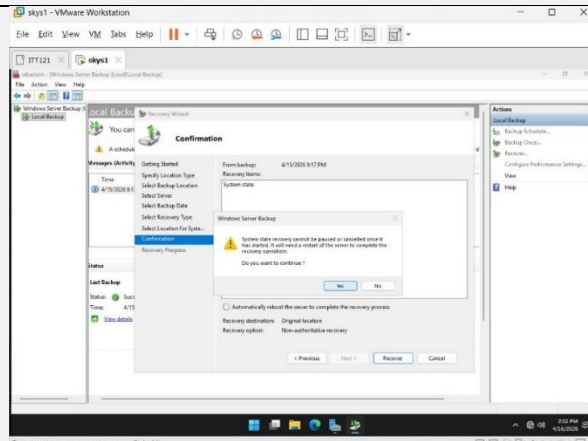


Backup and Restore

In the select location, select the option for “original location”. And click next.

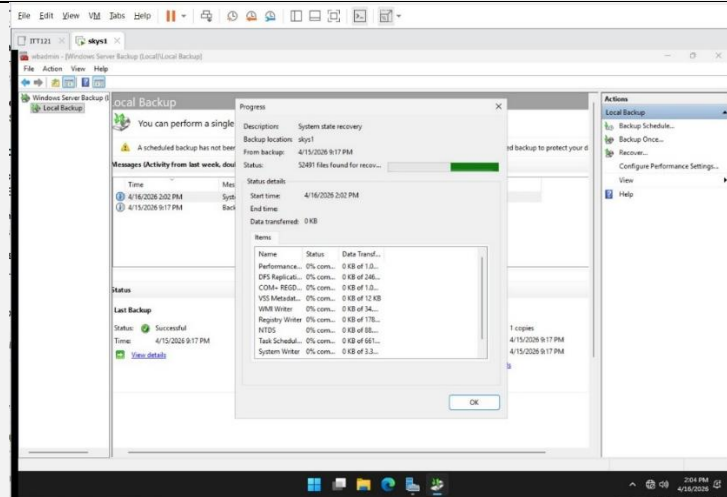


You will get a warning asking if you want to continue, click yes.

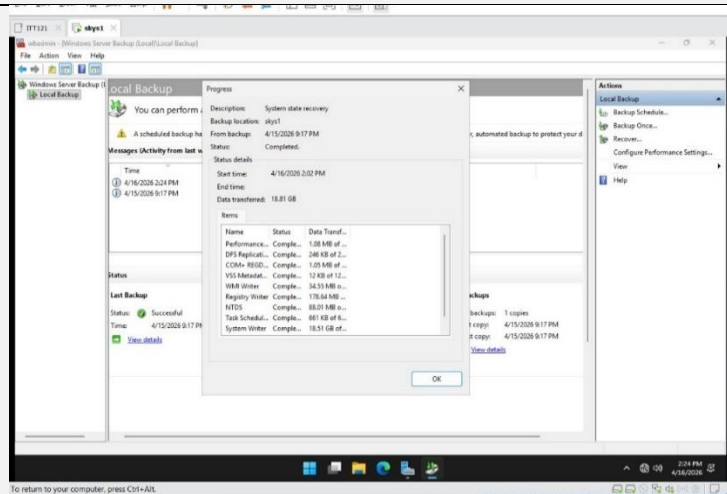


Backup and Restore

Now you can finish the setup and start the restore. You can see the progress and watch it.

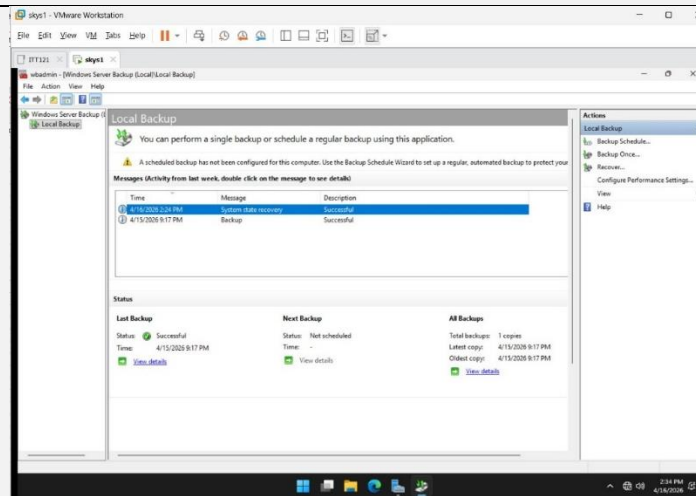


Once it is done backing up you can click ok.

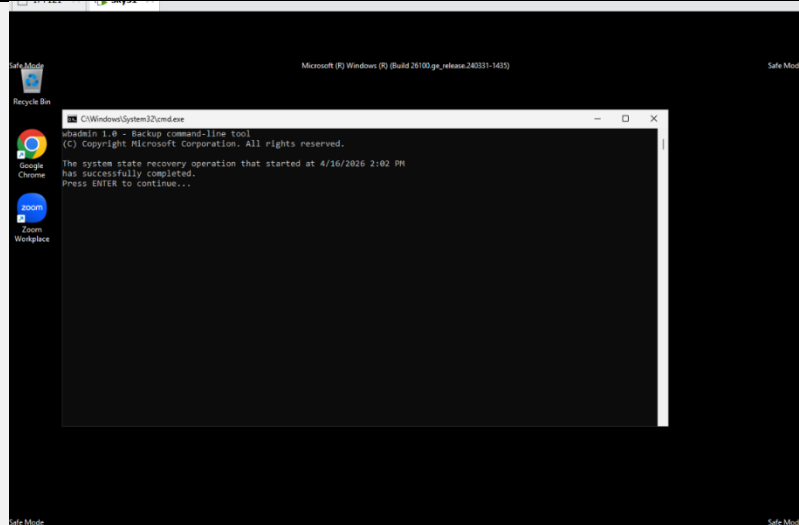


Backup and Restore

You should see the message
“successful” next to the
restore.

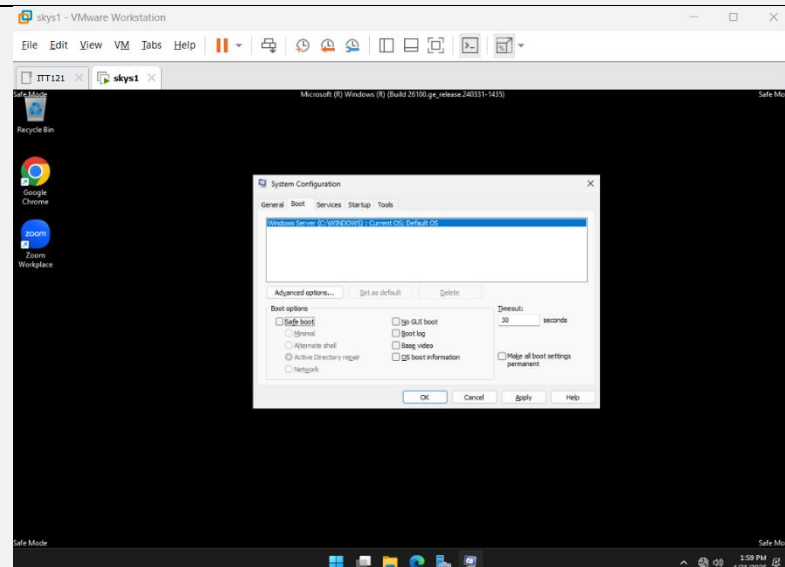


If you restart, command prompt will launch and let you know that the restore has successfully completed.



Backup and Restore

Now we need to turn off safe mode. In the start menu type out “msconfig” and hit enter. Go to the boot tab and disable safe mode. When you apply and close this application, it will ask you to restart, you can click restart.



When you sign into the server again, the accounting folder will be back in Active Directory Users and Computers.

